

1. 5 Figure 6.5 is the velocity-time graph for a train travelling north along a straight track.

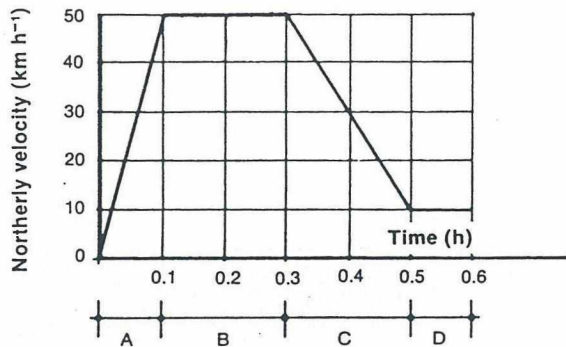


Fig. 6.5

- (a) In which section(s) (A, B, C, D)
- is the velocity a maximum?
 - is the acceleration a maximum?
 - are the velocity and acceleration constant throughout?
- (b) What distance is travelled by the train in section C?
- (c) What is the average speed of the train during the first 0.3 h? Give the answer to two significant figures.
- (d) Determine the acceleration of the train at 0.4 h. Give the answer to two significant figures.
3. 15 A cyclist commences a ride with a uniform acceleration of 1.0 m s^{-2} which he maintains for 8.0 s and then continues with a uniform speed. He pulls up with a uniform retardation over an interval of 13.5 s. If he travels a total distance of 438 m, for what period of time does he travel with a uniform speed?
5. 20 A motor-cycle patrolman has his machine stationary with the engine running when a speeding motor-car passes him at a speed of 126 km h^{-1} . As the car passes, he sets off after it with an acceleration of 1.40 m s^{-2} which he maintains until he overtakes it. Calculate the time required for the motor-cycle to overtake the car, and the distance travelled in doing this.
8. 7 A bushwalker travels 6.5 km north east, then 4.5 km north. What is his resultant displacement?
9. 9 In a tennis match at Kooyong a girl receives a serve which is travelling at 30 m s^{-1} south just prior to hitting her racquet. Immediately after leaving her racquet it is travelling at 25 m s^{-1} north. Specify fully the change in velocity she gave the ball.
12. 15 A ramp rises vertically 3.0 m for every 5.0 m of its length. A body weighing 100 N is placed on the ramp. Find the component of its weight
- parallel to the ramp;
 - perpendicular to the ramp.

2.

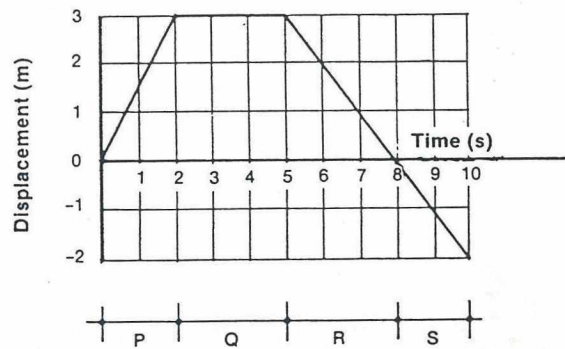


Fig. 6.10

- (a) What is the displacement of the object after 5 s?
- (b) Which section of the graph represents a constant velocity of 1.5 m s^{-1} ?
- (c) Which section of the graph represents the object at rest?
- (d) What was the instantaneous velocity of the object after 7 s?
- (e) What time elapsed before the object returned to its starting point?
4. 17 Between two stops a tram accelerates uniformly at the rate of 0.750 m s^{-2} for 12.0 s, travels for the next 20.0 s with the speed acquired, then comes to rest with uniform retardation which takes place over a distance of 27.0 m. Calculate the distance between the stops and the time taken for the tram to travel that distance.
6. 3 Two boys are dragging an object along level ground by pulling on a rope held horizontally with forces of 150 newton and 180 newton respectively. What single force on the rope would have the same effect?
7. 4 If the frictional force in the previous problem is 30 newton, what is the resultant horizontal force on the object?
10. 10 A cyclist travelling at 12 km h^{-1} east makes a right-hand turn at an intersection without changing speed. What is his change in velocity?
11. 11 A student whirls a rubber stopper on the end of a nylon string in a horizontal clockwise circle (as seen from above) at a constant speed of 6.0 m s^{-1} . From an instant when the stopper is moving in a northerly direction, find its change in velocity after moving round
- one-half of the circumference;
 - one complete turn;
 - one-quarter of a turn;
 - one-ninth of a turn.
13. 18 The air-speed of an aeroplane is 240 km h^{-1} and its course is north. It is carried out of its course by a wind blowing at 70.0 km h^{-1} from the west. Find its ground-speed and track.

ANSWERS.

1. (a) (i) B (ii) A (iii) B, D
(b) 6.0 km
(c) 42 km h^{-1}
(d) $-2.0 \times 10^2 \text{ km h}^{-2}$
2. (a) 3 m (b) P (c) Q
(d) -1.0 m s^{-1} (e) 8.0 s
3. 44 s
4. 261 m, 38.0 s
5. 50.0 s, $1.75 \times 10^3 \text{ m}$
6. 330 N pull
7. 300 N in direction of pull
8. $10.0 \text{ km N } 27^\circ \text{ E}$
9. $55 \text{ m s}^{-1} \text{ N}$
10. $17 \text{ km h}^{-1} \text{ SW}$
11. (a) $12 \text{ m s}^{-1} \text{ S}$ (c) $8.5 \text{ m s}^{-1} \text{ SE}$
(b) 2000 (d) $4.1 \text{ m s}^{-1} \text{ E } 20^\circ \text{ S}$
12. (a) 60 N
(b) 80 N
13. $250 \text{ km h}^{-1} \text{ N } 16^\circ \text{ E}$

16 A man pushes a garden roller weighing 1000 N over a level lawn, holding the end of a 150 cm long handle 90 cm above the axis of the roller as shown in Figure 8.3. He pushes down the handle with a force of 200 N.

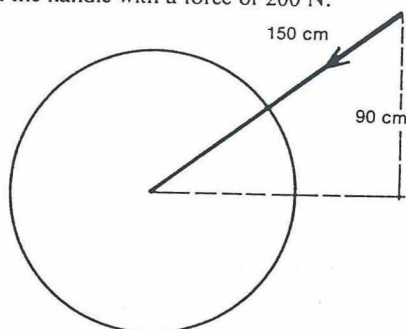


Fig. 8.3

- What is the size of the effective force pushing the roller along?
- With what force does the roller press against the lawn while the man is pushing the handle?
- What force would the roller exert on the lawn if the man pulled it (with the same force) instead of pushing it, the handle remaining in the same position?
- If the roller lodges in a small rut, would it be easier to get it out by pulling or by pushing?

21. 13 A net force of 20.0 N acts on a 5.00 kg mass for 6.00 s. The original velocity of the mass is 100 m s^{-1} east. Calculate the velocity after the 6.00 s, if
- the force is applied in an easterly direction;
 - the force is applied in a westerly direction.

22. 14 What is the momentum of
- a 70 kg sprinter running north at 9.0 m s^{-1} ?
 - an apple of mass 110 g falling downwards at 5.0 m s^{-1} ?

25. 31 A net force of 450 N acts for 4.00 s on a body of mass 90 kg. Calculate the acceleration, the speed acquired, and the distance travelled in the 4.00 s.

26. 32 A proton, of mass $1.7 \times 10^{-27} \text{ kg}$, moving at a speed of $5.0 \times 10^7 \text{ m s}^{-1}$, is brought to rest by a constant force in a distance of 0.25 m. What is the size of the force?

27. 6 A bucket of water with a total mass of 20 kg is raised from a well by means of a vertical rope. During the first part of the rise the bucket has an upward acceleration of 0.200 m s^{-2} . During this initial accelerating period,

- what is the net force acting on the bucket of water?
- what is the force exerted by the rope on the bucket of water?

29. 10 A mass of 3.00 kg is hung from a spring balance (graduated in newtons) in a lift. Calculate the reading of the spring balance while the lift is slowing to a stop on an upward trip, the magnitude of the acceleration being 2.00 m s^{-2} .

- 19 A yacht travelling with a velocity of 10 km h^{-1} north relative to the water is in a tidal stream flowing south-east at 4.0 km h^{-1} . Determine, either by scale drawing or by calculation, the yacht's velocity relative to the ocean floor.

- 20 An aeroplane travels due east at 360 km h^{-1} relative to the ground while the wind is blowing from the south at 100 km h^{-1} . In what direction does the pilot point the plane, and what is the speed of the plane through the air?

- 21 A motor-cyclist is travelling due north with a speed of 60.0 km h^{-1} and a wind is blowing from the east at 20.0 km h^{-1} . What is the apparent direction and speed of the wind to the cyclist?

- 25 A river 2.0 km across has a current of velocity 8.0 km h^{-1} . In what direction must a launch, moving at 16 km h^{-1} relative to the water, steer in order to reach one bank directly opposite its starting point on the other? What will be the time for the crossing?

19. 10 What is the magnitude of the force needed to accelerate an electron of mass $9.1 \times 10^{-31} \text{ kg}$ from rest to a speed of 10^5 m s^{-1} in 10 s?

20. 12 A cricket ball, of mass 0.14 kg, moving horizontally at 24 m s^{-1} north was hit back with a speed of 16 m s^{-1} south. If the contact with the bat lasted for $4.0 \times 10^{-2} \text{ s}$ find the average force exerted by the bat on the ball.

23. 15 (a) Find the impulse given to a body by a constant force of (i) 6.0 N upwards acting for 0.50 s; (ii) 40 N south-west acting for 10 s.
- (b) What would be the change in momentum of the bodies in each of the cases (i) and (ii) above?

24. 16 A net force of 20 N towards the east acts on a mass of 8.0 kg for a period of 5.0 s. Find
- the impulse of the force;
 - the change in momentum of the body;
 - the final momentum of the body if its momentum when the force began to act was 20 kg m s^{-1} west;
 - the final velocity of the body.

28. 9 The mass of a lift and its occupants is 600 kg. Calculate the tension in the cable supporting the lift when it is moving
- upwards with a uniform velocity;
 - with a uniform acceleration of 0.450 m s^{-2} directed downwards;
 - with a uniform acceleration of 0.450 m s^{-2} directed upwards.
- (d) Can you tell from the statements in (b) and (c) above whether the lift is moving upwards or downwards in each case?

30. 11 A man weighing 980 N slides down a rope that can support a weight of only 755 N.
- How is this possible?
 - What is the least acceleration he can have without breaking the rope?
 - What will be his minimum speed after sliding down 8.0 m?

ANSWERS.

14. (a) $1.6 \times 10^3 \text{ N}$ (b) $1.1 \times 10^3 \text{ N}$ (c) $8.8 \times 10^2 \text{ N}$
(d) pulling

15. 7.8 km h^{-1} N 22° E

16. 374 km h^{-1} E 15.5° S

17. 63.2 km h^{-1} N 18.4° E

18. 60° to bank upstream, 8.7 min

19. $9.1 \times 10^{-27} \text{ N}$

20. $1.4 \times 10^2 \text{ N}$ South

21. (a) 124 m s^{-1} E (b) 76.1 m s^{-1} E

22. (a) $6.3 \times 10^2 \text{ kg m s}^{-1}$ N

(b) 0.55 kg m s^{-1} down

23. (a) (i) 3.0 N s up (ii) $4.0 \times 10^2 \text{ N s}$ SW

(b) (i) 3.0 N s up (ii) $4.0 \times 10^2 \text{ N s}$ SW

24. (a) $1.0 \times 10^2 \text{ N s}$ E (b) $1.0 \times 10^2 \text{ kg m s}^{-1}$ E

(c) 80 kg m s^{-1} E (d) 10 m s^{-1} E

25. 5.0 m s^{-2} , 20.0 m s^{-1}
40 m

26. $8.5 \times 10^{-12} \text{ N}$

27. (a) 4.00 N up
(b) 200 N up

28. (a) $5.98 \times 10^3 \text{ N}$
(b) $5.61 \times 10^3 \text{ N}$
(c) $6.15 \times 10^3 \text{ N}$
(d) NO.

29. 23.4 N (b) 2.25

31. 2 A stone is dropped from a height of 51.2 m above the ground. (Take $g = 10 \text{ m s}^{-2}$.)
 (a) How long does it take to reach the ground?
 (b) What is its speed just before it strikes the ground?
 (c) How far does it fall during the last second of its motion?

32. 3 A balloon is moving vertically downwards with a uniform speed of 5.0 m s^{-1} . When it is 60 m from the ground, a small object is dropped from it. How long before the balloon will the small object reach the ground? (Take $g = 10 \text{ m s}^{-2}$.)

- ~~35.~~ 8 A stone is thrown with a velocity of 20 m s^{-1} at an angle of 40° with the horizontal. Determine
 (a) the time for which it is above the level at which it is thrown;
 (b) the greatest vertical height reached;
 (c) its range on a horizontal plane.

37. 11 A body of mass 4.0 kg is moving at a constant velocity of 3.0 m s^{-1} west. It is then acted upon by a force of 5.0 N west, while it moves 2.0 m west.
 (a) What is the work done by the force?
 (b) What is the resulting increase in the kinetic energy of the body?
 (c) What was the initial kinetic energy of the body?
 (d) What is the final kinetic energy of the body?
 (e) Determine its final speed.

40. 23 The engine of a jet aeroplane develops 2.4 megawatt of power when in level flight at a constant speed of 200 m s^{-1} . Calculate the air resistance on the plane.

- ~~41.~~ 24 Find the power necessary to move a body of mass 1.5 kg up a smooth plane inclined at 30° to the horizontal with a uniform speed of 5.0 m s^{-1} .

43. 7 A pendulum bob is pulled aside until it is 10 cm vertically above its lowest point and then released. With what speed does it pass through its mean position?

33. 9 A balloon is 30.0 m above the ground and is rising vertically with a uniform speed when a coin is dropped from it. If the coin reaches the ground in 4.00 s, what is the speed of the balloon?

- ~~34.~~ 3 A ball is thrown horizontally with a speed of 14.0 m s^{-1} from a point 6.40 m above the ground. Calculate
 (a) the time taken by the ball to reach ground level;
 (b) the horizontal distance travelled in that time;
 (c) its velocity when it reaches the ground.

- ~~36.~~ 11 A cricketer hits a six, the ball having travelled 100 m horizontally when it strikes the ground beyond the boundary. The greatest height reached by the ball is 25 m. Calculate the velocity with which the ball left the bat.

38. 15 A body of mass 50 kg moving with a speed of 20 m s^{-1} is brought to rest by a constant force in a distance of 5.0 m. Calculate
 (a) the loss in kinetic energy of the body;
 (b) the work done by the body against the retarding force;
 (c) the size of the force.

39. 21 An upward force of 6.0 kN raises a mine cage a vertical distance of 100 m in 40 s at a constant rate.
 (a) How much work is done by the force in the 40 s?
 (b) What work is done by the force in one second?
 (c) At what power is the force working?

42. 5 A boy on a bridge drops a 5.0 kg rock into the water 10 m below. Taking the water surface as the zero gravitational potential energy level, determine
 (a) the gravitational potential energy of the rock as it left the boy's hand;
 (b) its total energy (kinetic plus potential) as it left his hand;
 (c) the total energy of the rock half-way down;
 (d) its kinetic energy just before hitting the water;
 (e) its speed just before hitting the water.

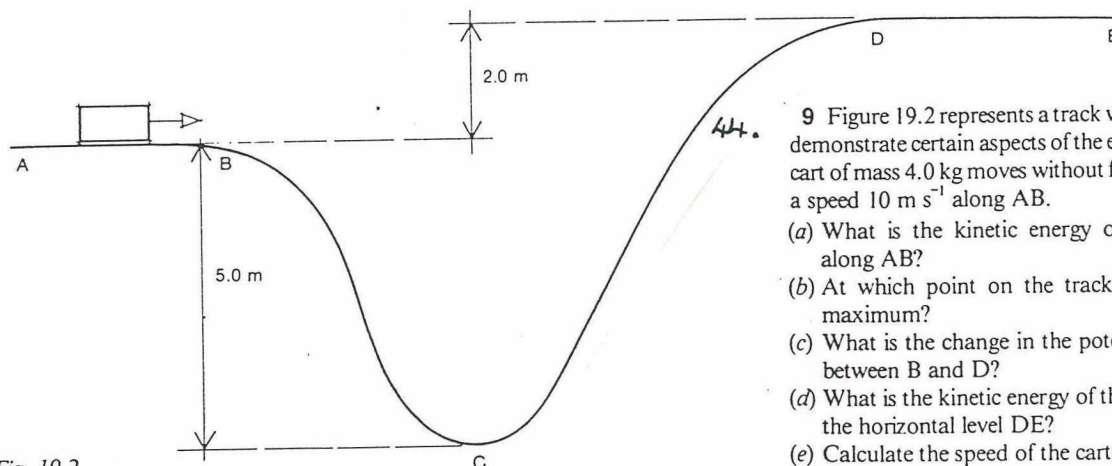


Fig. 19.2

44. 9 Figure 19.2 represents a track which has been erected to demonstrate certain aspects of the energy of a body. A small cart of mass 4.0 kg moves without friction on the track, with a speed 10 m s^{-1} along AB.

- (a) What is the kinetic energy of the cart as it moves along AB?
 (b) At which point on the track is its kinetic energy a maximum?
 (c) What is the change in the potential energy of the cart between B and D?
 (d) What is the kinetic energy of the cart as it moves along the horizontal level DE?
 (e) Calculate the speed of the cart as it passes the point C.

ANSWERS.

31. (a) 3.2 s (b) 32 m s^{-1} (c) 2.7 m

32. 9.0 s

33. 12.1 m s^{-1}

34. (a) 1.14 s (b) 16.0 m
 (c) 17.9 m s^{-1} 38.7° to horizontal

35. (a) 2.6 s (b) 8.4 m (c) 40 m
 36. 31 m s^{-1} at 45° to horizontal

37. (a) 10 J (b) 10 J (c) 18 J (d) 28 J
 (e) 3.7 m s^{-1}

38. (a) $1 \times 10^4 \text{ J}$ (b) $1 \times 10^4 \text{ J}$ (c) $2.0 \times 10^3 \text{ N}$

39. (a) $6.0 \times 10^5 \text{ J}$ (b) $1.5 \times 10^4 \text{ J}$ (c) $1.5 \times 10^4 \text{ W}$

40. $1.2 \times 10^4 \text{ N}$

41. 37 W

42. (a) $4.9 \times 10^2 \text{ J}$ (b) $4.9 \times 10^2 \text{ J}$ (c) $4.9 \times 10^2 \text{ J}$
 (d) $4.9 \times 10^2 \text{ J}$ (e) 14 m s^{-1}

43. 1.4 m s^{-1}

44. (a) $2 \times 10^2 \text{ J}$ (b) C (c) 78 J (d) $1.2 \times 10^2 \text{ J}$
 (e) 14 m s^{-1}

1 Figure 30.1 illustrates a pulse which is moving along a stretched spring at a speed of 0.50 m s^{-1} .

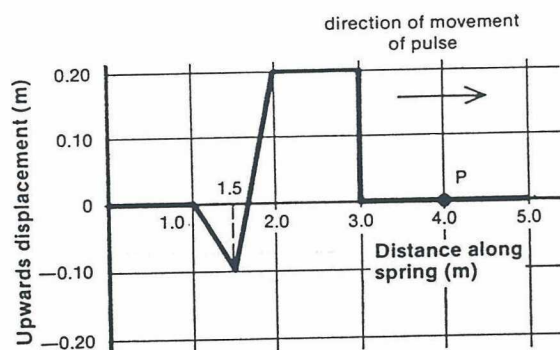


Fig. 30.1

- (a) Draw a displacement–time graph for the particle P (4.0 m from the origin), taking zero time at the instant for which the graph in Figure 30.1 was drawn.
- (b) What will be the velocity of the particle P after 3 s, 4.5 s and 5.5 s respectively?
- (c) You will appreciate that the graph in Figure 30.1 is an idealisation. Give one reason why an actual pulse could not have the shape shown.

11 Determine the size, nature and position of the image formed when an object 3.0 cm high is placed 20 cm from a convex mirror of radius of curvature 60 cm.

13 A convex mirror and a plane mirror are placed facing each other and 28 cm apart. Midway between them and on the principal axis of the convex mirror a small luminous object is placed. If the distance between the two images observed on looking in the plane mirror is 24 cm, calculate the radius of curvature of the convex mirror.

5 A faulty barometer has some air trapped above the mercury. The height of the mercury column is 732 mm when the air space is 100 mm long. When the tube is pushed down into the bowl so that the air space is 50.0 mm long, the height of the mercury column is 716 mm. What is the correct atmospheric pressure

- (a) in mmHg?
- (b) in kPa?

8 A quantity of gas has a volume of 120 cm^3 at a temperature of 7°C . What would be its volume at 77°C , the pressure remaining constant?

18 If $4.2 \times 10^3 \text{ J}$ of heat energy are required to vaporise 5.0 g of ethanol at its boiling point, what is the specific latent heat of vaporisation of ethanol?

3 A guitar string is plucked, sending 50 progressive waves along it in 0.200 s. What is

- (a) the frequency, and
- (b) the period of the waves?

6 Longitudinal pulses of frequency 2.4 Hz are sent along a slinky spring on the floor at a speed of 3.6 m s^{-1} . What is their wavelength?

2 Determine the size, nature and position of the image formed when an object 2.0 cm high is placed 15 cm from a concave mirror of focal length 10 cm.

3 When an object is placed 40 cm from a concave spherical mirror, the image formed is also 40 cm from the mirror.

- (a) Is the image inverted or erect?
- (b) Is it real or virtual?
- (c) What is its magnification?
- (d) What is the focal length of the mirror?
- (e) Is the radius of curvature of the mirror about 10, 20, 40 or 80 cm?

7 A concave spherical mirror has a focal length 8.0 cm. An object is placed in such a position that there is formed

- (a) a real image of half the height of the object;
- (b) a virtual image of twice the height of the object.

Find the distances of the object and image from the mirror in each case.

8 A dentist uses a concave mirror for examining patients' teeth. Design a mirror suitable for this purpose, assuming that the image is to be three times the size of the object, erect, and that the mirror is not to be held nearer to the tooth than 1.0 cm.

15 A pin is set up 30 cm in front of a convex mirror. It is found that if a plane mirror is placed between the pin and the convex mirror 20 cm from the pin (so that it covers only half the convex mirror) the images of the pin in both convex and plane mirrors coincide. Find the focal length of the convex mirror.

7 A 120 m^3 balloon to carry cosmic ray experiments to the upper atmosphere is to be filled with hydrogen at atmospheric pressure (100 kPa). If the hydrogen comes in gas cylinders with an internal volume of 0.042 m^3 and an internal pressure of 20 MPa, how many such cylinders will be needed? Assume the volume of the balloon remains effectively constant as it rises.

19 A 2100 W electric kettle containing 1.00 litre (i.e. 1.00 kg) of water has just come to the boil.

- (a) How much energy is required to boil all the water away?
- (b) How long (to the nearest minute) will it take for the kettle to boil dry?

ANSWERS.

45. (b) zero, 0.70 m s^{-1} downwards, 0.10 m s^{-1} upwards.

46. (a) 250 Hz (b) $4.00 \times 10^{-3} \text{ s}$.

47. 1.5 m

48. real, inverted, 4.0 cm high

49. (a) inverted (b) real (c) 1.0 (d) 20 cm
(e) $\sim 40 \text{ cm}$.

50. (a) object: 24 cm image: 12 cm
(b) $u = 4.0 \text{ cm}$ $v = 8.0 \text{ cm}$

51. $f = 1.5 \text{ cm}$

52. virtual, upright, 1.8 cm high, 12 cm behind the mirror.

53. 70.0 cm

54. 15 cm

55. (a) 748 mm Hg (b) 99.5 kPa

56. 15 cylinders

57. 150 cm^3

58. $8.4 \times 10^5 \text{ J kg}^{-1}$

59. (a) $2.26 \times 10^6 \text{ J}$ (b) 18 min.

60. 21 A cross-country skier descends a 25° slope at a constant speed of 12 m s^{-1} . If all the work done by friction on the lower side of her skis goes into producing heat, what mass of snow does she melt in the 50 s she is descending the slope? Her total mass including skis and pack is 80 kg.

62. 26 A copper calorimeter has a mass of 50 g and contains 85 g of water at 16°C . 6.0 g of dried ice is then added and the contents stirred until all the ice has melted. What is the temperature of the mixture? Specific heat capacity of copper = $3.8 \times 10^2 \text{ J kg}^{-1} \text{ K}^{-1}$.

63. 29 A calorimeter which has a mass of 1.00 kg and a specific heat capacity of $8.4 \times 10^2 \text{ J kg}^{-1} \text{ K}^{-1}$ is cooled to 0°C and 1.40 kg of ice at 0°C is placed in it. 1.00 kg of water at 100°C is then poured into the calorimeter. Find the resulting temperature.

64. 5 What is the force of repulsion between two point charges of $+6 \mu\text{C}$ (microcoulomb) and $+5 \mu\text{C}$, situated 0.3 m apart in a vacuum?

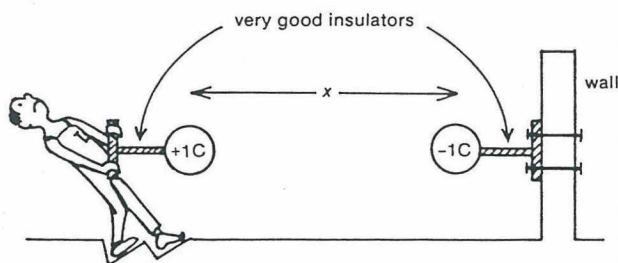


Fig. 54.4

65. 6 In order to appreciate the force of attraction between two objects carrying charges of $+1 \text{ C}$ and -1 C , imagine the situation in Figure 54.4. With your feet well dug in, you could probably exert a horizontal force of, say, 560 N. Guess an answer for the minimum value of x , and then actually calculate it. You might be surprised.

67. 13 What is the potential difference between two points, if
(a) 24 J of work is done to move a total charge of 3 C from one point to the other?
(b) an alpha particle (charge $3.2 \times 10^{-19} \text{ C}$) gains $1.6 \times 10^{-15} \text{ J}$ of energy in moving freely between two points?

68. 14 How much work is done by an electric field which moves a total charge of 5.0 C from one end of the heating element in an electric jug to the other, if the potential difference between the two ends is 240 V?

71. 32 A d.c. power pack with an e.m.f. of 8.0 V is delivering a current of 2.5 A.
(a) What is the power output of the power pack?
(b) If it is used for 5.0 min, how much electrical energy does it supply?

61. 25 In Figure 49.2 a hypothetical temperature–time curve has been drawn for 0.60 kg of a solid which is warmed in a well-insulated container by a 100 W heater.

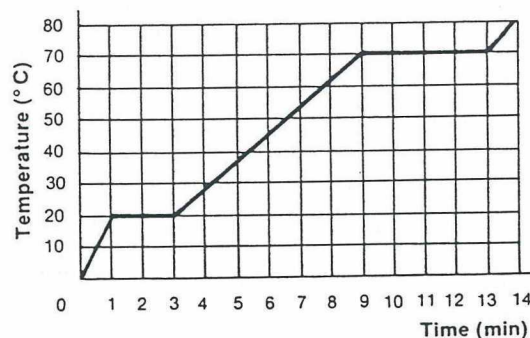


Fig. 49.2

- (a) What is the melting point of the solid?
(b) Which is greater: the substance's specific latent heat of fusion or vaporisation?
(c) Which is greatest: the substance's specific heat capacity as a solid, liquid or gas?
(d) Determine its specific latent heat of fusion.
(e) What is its specific heat capacity as a liquid?

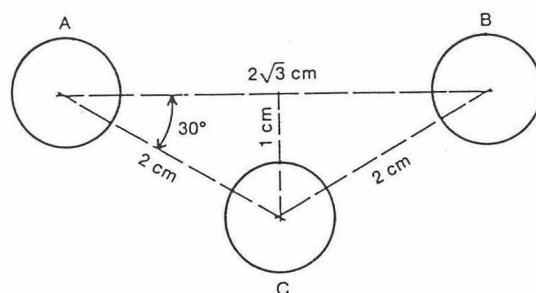


Fig. 54.8

66. 11 In Figure 54.8, A, B and C are three small insulated conductors all carrying positive charges of equal size. A exerts a force of $2.5 \mu\text{N}$ on B.

Give answers to two significant figures.

- (a) What is the magnitude of the force exerted by B on C?
(b) What is the direction of the resultant force exerted by A and B on C?
(c) What is the magnitude of this resultant force?

69. 26 There is a current of 5.0 A in the heating element of a portable refrigerator. How long does it take for 60 C of charge to flow through the heating element?

70. 27 In a particular computer monitor the current in the electron beam between the cathode and the anode is 2.5 mA. In 32 s

- (a) what charge, and
(b) how many electrons leave the cathode?

ANSWERS.

60. 0.60 kg

61. (a) 20°C (b) L.v (c) heat capacity as a liquid
(d) 20 kJ kg^{-1} (e) $1.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$

62. 10°C

63. 0°C

64. 3 N.

65. 4 km

66. (a) $7.5 \times 10^{-6} \text{ N}$ (b) down (c) $7.5 \times 10^{-6} \text{ N}$

67. (a) 8 V (b) $5.0 \times 10^3 \text{ V}$

68. $1.2 \times 10^3 \text{ J}$

69. 12 s

70. (a) 0.080 C (b) 5.0×10^{17} electrons

71. (a) 20 W (b) $6.0 \times 10^3 \text{ J}$

72. 14 A common type of copper wire has a cross-sectional area of $1.00 \times 10^{-8} \text{ m}^2$. If a 6.03 m length of it has a resistance of 10.2Ω , what is the resistivity of copper?
73. 15 If a standard resistor of 6.00Ω is to be made of constantan wire of diameter 0.250 mm, what length is required? Resistivity of constantan = $4.9 \times 10^{-7} \Omega \text{ m}$.
75. 40 What is the effective total resistance between the points P and Q in
(a) Figure 56.14?
(b) Figure 56.15?

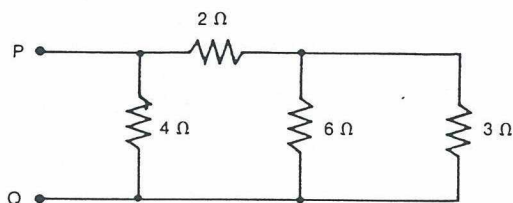


Fig. 56.14

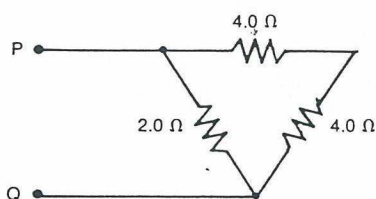


Fig. 56.15

76. 41 Assume you have only five 12Ω resistors. How could you connect some or all of them to produce an effective total resistance of
(a) 48Ω ?
(b) 4Ω ?
(c) 15Ω ?
(d) 28Ω ?
78. 22 Resistors having 2, 3, 4 and $R \Omega$ resistance are arranged as in Figure 57.12, there being a potential difference of 2 V applied between A and C.

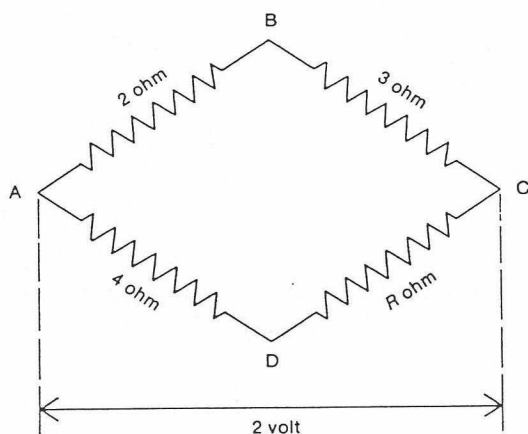


Fig. 57.12

- Answers to (a) and (b) may contain R .
(a) What is the current in each of the arms AB, BC, AD and DC?
(b) What is the potential difference between
(i) A and B?
(ii) A and D?
(c) Use your answers to (b) to determine a value for R for which a galvanometer connected between B and D would read zero.

ANSWERS.

72. $1.69 \times 10^{-8} \Omega \text{ m}$

73. 0.60 m

74. (a) $8.4 \times 10^3 \text{ J}$ (b) 20 W (c) 8.0 V

75. (a) 2Ω (b) 1.4Ω

76. (a) 4 in series (b) 3 in parallel

(c) 4 in parallel, 1 in series

(d) 3 in parallel, 2 in series.

77. (a) 9.0 V (b) 1.0 A (c) 1.5 A (d) 3.0 V

74. 25 A well-insulated calorimeter contains 0.20 kg of water at 15°C . A heating coil R is placed in the water and connected to a power source as indicated in Figure 56.9. The temperature of the water rises to 25°C in 7.0 min while the ammeter reads 2.5 A. Assume the heat lost to the calorimeter and the surroundings is negligible. Specific heat capacity of water = $4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

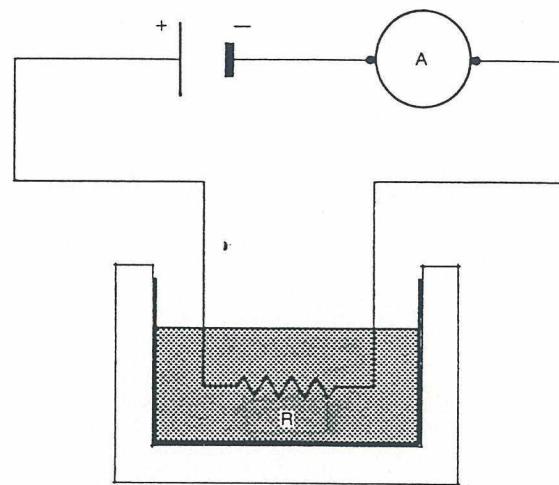


Fig. 56.9

Determine

- (a) the amount of electrical energy transformed into heat energy in the 7.0 min;
(b) the power developed in R;
(c) the potential difference across R.
77. 45 A circuit is shown in Figure 56.17, in which the straight wires represent conductors of negligible resistance.

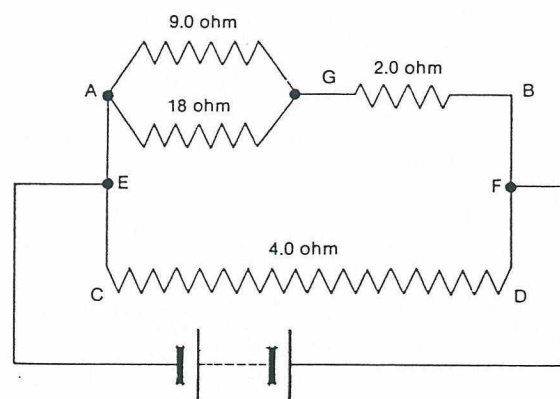


Fig. 56.17

- If the current in the 18Ω resistor is 0.50 A , what is
(a) the potential difference between A and G?
(b) the current in the 9.0Ω resistor?
(c) the current in the 2.0Ω resistor?
(d) the potential difference between G and B?
(e) the potential difference between A and B?
(f) the potential difference between C and D?
(g) the current in the 4.0Ω resistor?
(h) the current in the battery?

78. (a) AB, BC: 0.4 A

AD, DC: $\frac{2}{4+R} \text{ A}$

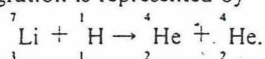
(b)(i) 0.8 V

(ii) $\frac{8}{4+R} \text{ V}$

79. 10. What is the magnitude of the charge on
 (a) an alpha-particle;
 (b) a deuteron;
 (c) a neutron;
 (d) an oxygen nucleus (atomic number 8)?

82. 28. When lithium 7 is bombarded with protons, two alpha-particles, possessing considerable kinetic energy, are produced.

The disintegration is represented by



Using the atomic masses listed below, calculate the kinetic energy possessed by the two alpha-particles formed when one lithium 7 nucleus disintegrates.

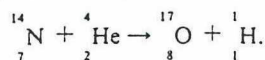
Lithium 7 7.01818 u

Proton 1.00813 u

Alpha-particle 4.00389 u

83. 29. When nitrogen 14 is bombarded with alpha-particles, the disintegration results in the formation of oxygen 17 and a proton. Is energy released or absorbed? Calculate (in eV) the change in energy associated with the disintegration of one nitrogen nucleus.

The disintegration is represented by



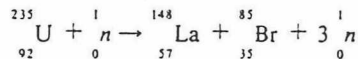
Nitrogen 14 14.00750 u

Oxygen 17 17.00450 u

Use values given for alpha-particle and proton in No. 28.

80. 27. Calculate the energy that would result from the conversion of 1.0 gram of mass into energy. Express the result in joule and in electron volt.

81. 30. A uranium 235 nucleus can fission in a number of ways. One case, in which lanthanum and bromine are produced, is represented as follows



Uranium 235 235.124 u

Neutron 1.009 u

Lanthanum 147.961 u

Bromine 85 84.938 u

Calculate the energy released in the fission of one uranium nucleus.

ANSWERS

79. (a) $+3.20 \times 10^{-19} \text{ C}$

(b) $+1.60 \times 10^{-19} \text{ C}$

(c) zero

(d) $+1.28 \times 10^{-18} \text{ C}$

80. $9.00 \times 10^{13} \text{ J}$, $5.60 \times 10^{32} \text{ eV}$

81. $3.20 \times 10^{-11} \text{ J}$

82. $2.80 \times 10^{-12} \text{ J}$

83. $1.20 \times 10^6 \text{ eV}$, absorbed.